

INTRODUCTION to BLOCKCHAIN

SYLLABUS

Dr. Noureddine Lasla

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Who am I?

Dr. Noureddine Lasla

- Associate Professor at ENISA (Since March 2024).
- Assistant Professor at ENISA (From June 2022 – Feb 2024).
- Research Fellow (HBKU, Qatar, 2017 - 2021).
- Research Fellow (CERIST, Algeria, 2009 - 2017).
- Ph.D. in Computer Science (USTHB, 2015).

Experience in Blockchain tech. since 2017

Research area: Distributed Systems, Blockchain, IoT, Cyber Security



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Course Overview

Objective:

To provide a comprehensive understanding of blockchain technology, from fundamental principles to advanced applications using smart contracts.

Key Areas of Focus:

- Blockchain fundamentals
- Cryptocurrencies
- Consensus mechanisms
- Smart contracts

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Course structure

Chapters:

1. Fundamentals of Blockchain
2. Technical Architecture and Basic Components
3. Cryptocurrency
4. Consensus Mechanisms
5. Smart Contracts

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Course structure (5 weeks)

Week	Chapters
1	**Fundamentals of Blockchain** • Introduction to blockchain: Concepts and purpose • Key characteristics: Decentralization, transparency, immutability, and security • Blockchain structure: Blocks, chains, nodes • Blockchain use cases: Cryptocurrency, supply chain, identity verification
2	**Technical Architecture and Basic Components** • Blockchain architecture: Nodes, miners, ledgers • Block structure: Merkle trees, hash functions • Public and private key cryptography • Data integrity and security mechanisms and attacks

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Course structure (5 weeks)

Week	Chapters
3	**Cryptocurrency:** • Introduction to cryptocurrency: Bitcoin, Ethereum, altcoins • Cryptocurrency wallets and transactions • Mining concepts: Proof of Work (PoW)
4	**Consensus Mechanisms:** • Consensus mechanisms: Proof of Work (PoW), Proof of Stake (PoS), Proof of History (PoH) • Byzantine Fault Tolerance (BFT) • Comparison of various consensus algorithms • Security and performance trade-offs
5	**Smart Contracts** • What are smart contracts? How do they work? • Smart contract platforms: Solidity, Ethereum • Use cases of smart contract

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Assessment and Grading

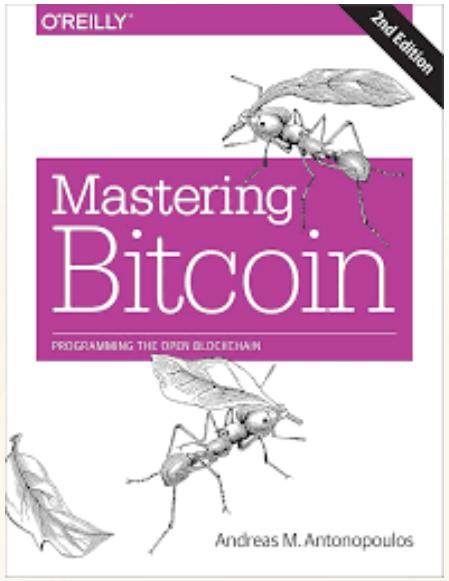


- Final Examination [**60 %**]
- Lab Test [**35%**]
- (CE) Attendance and participation in the lab [**5%**]

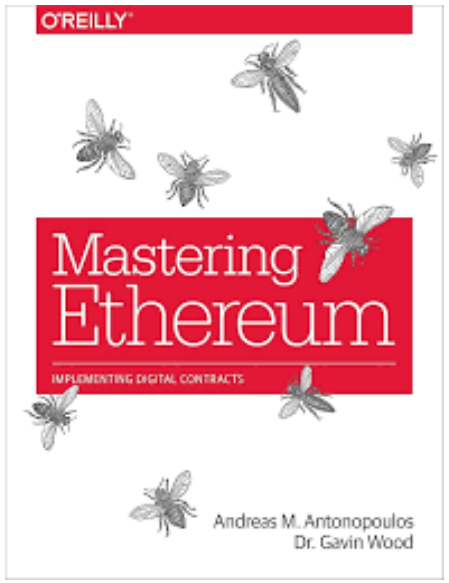
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References

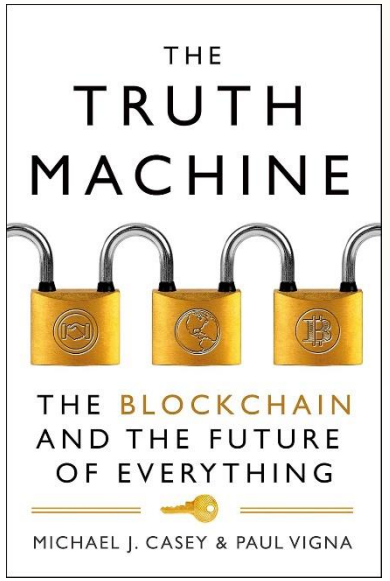
Textbooks:



Mastering Bitcoin: Programming the Open Blockchain
by Andreas M. Antonopoulos
2nd Edition



Mastering Ethereum: Building Smart Contracts and Dapps
by Andreas M. Antonopoulos



The Truth Machine: The Blockchain and the Future of Everything
by Paul Vigna, Michael J. Casey

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